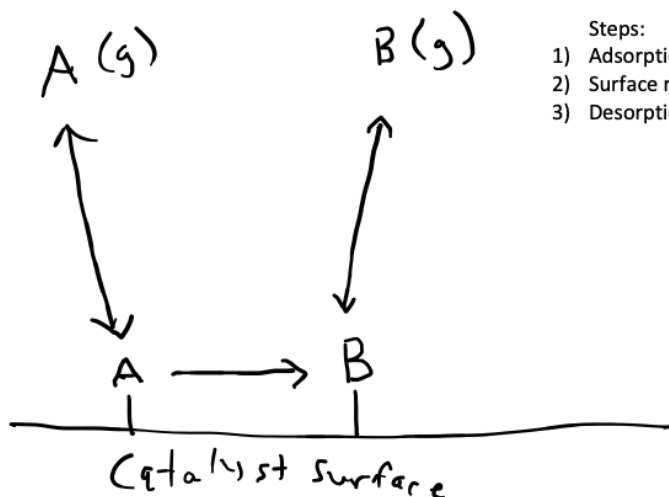


32 Mechanisms in Heterogeneous Catalysis

Consider the reaction $A(g) \rightarrow B(g)$, which is happening over a catalyst:

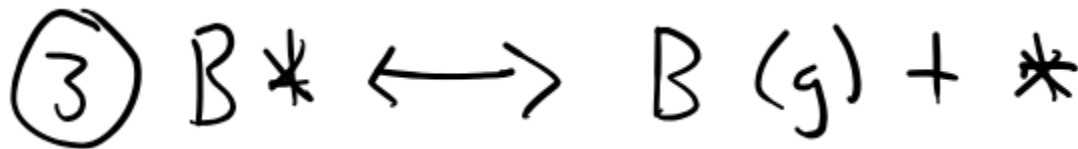
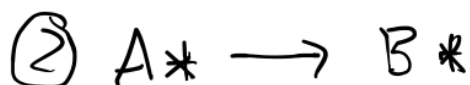
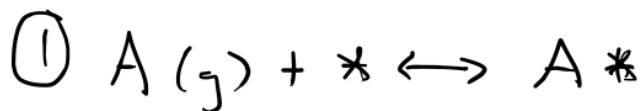


- Steps:
- 1) Adsorption of A
 - 2) Surface reaction: $A \rightarrow B$
 - 3) Desorption of B

mechanisms are series of elementary steps

Reaction "mechanism": Series of steps taken to get from reactants to products.

Mechanism in this example:



* are catalytic sites

*ed species are bound to the catalyst

we can think of $A*$ as $A-Pt$ if the catalyst is Pt

Adsorption and desorption steps are often "unactivated." This means that there is no activation barrier. Adsorption & desorption steps are driven by thermodynamics & not kinetics.

The surface reaction is thus often the "rate limiting step" or "rate determining step" (RDS)

Rate limiting step:

is the slowest step in the mechanism, i.e., the bottleneck step. Defines the rate of reaction. Because of this, we assume that the RLS/RDS is the only slow step in the mechanism, all other steps are thus @ equilibrium ($v_{net} = 0$).

Deriving mechanisms for catalytic reactions:

- It is useful to know the reaction mechanism and rate limiting step, because it allows you to tune the catalyst properties to carry out the reaction more efficiently
- One way to estimate the mechanism is to run an experiment to get the rate law, and then devise a mechanism and rate limiting step that explain the experimental observations
- Actual research example: $2 \text{ NO (g)} + \text{O}_2 \text{ (g)} \rightarrow 2 \text{ NO}_2 \text{ (g)}$ over a Pt catalyst

NOTE: This is an actual research example that demonstrates "exceptions to the rule" and complexities that are usually intentionally omitted from textbook and exam problems.

Observed rate law: $-r_{\text{NO}} = \frac{k C_{\text{NO}} C_{\text{O}_2}}{C_{\text{NO}_2}}$, i.e., 1st order in NO and O₂ and -1st order in NO₂

We want to devise a mechanism and rate limiting step that explain this behavior. What steps could we/should we include?

← point out here that O* is the MASI & all other species except maybe NO have coverages << θ*

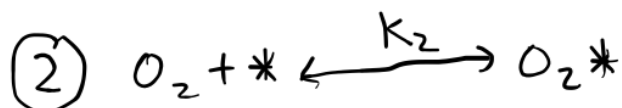
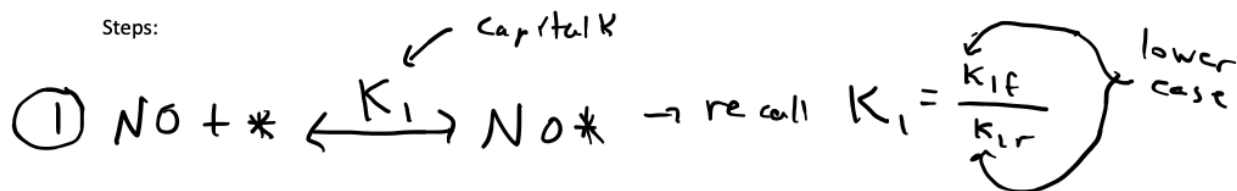
NO adsorbs to the catalyst

O₂ adsorbs to the catalyst

O₂ dissociates on the catalyst to form 2 O*

$\text{NO}^* \frac{1}{2} \text{O}^*$ react to form NO_2^*
 NO_2 desorbs from the catalyst

Steps:



Note: "sites," i.e., combinations of *'s and *'ed species, must balance in each step.

How do we get the rate from this mechanism?

The rate of the reaction $\approx$ the rate of the rate limiting step. $\leftarrow$ assumption/model

Which step should we assume is the rate limiting step?

Let's choose step ③ $\leftarrow$ let students pick $\leftarrow$ point out that all other steps are $\approx$ temporary equilibrium as they wait for the overall rxn to achieve equilibrium

If step 3 is the rate limiting step, then all other steps (including step 4) are assumed to be in equilibrium. i.e., the net rates of all other steps are equal to 0.

Under our assumption, the rate of NO consumption = $-r_{NO}$ = forward rate of step 3
(so here we've assumed that step 3 is irreversible)

Therefore, $-r_{NO} = r_3 = k_3 \theta_{O_2} \theta_*$

Here θ_i are "coverages," i.e., $\frac{\text{\# of catalytic species occupying a catalytic site}}{\text{\# of catalytic sites}} = \frac{\text{\# of catalytic species}}{\text{catalyst surface area}}$

By definition, $0 \leq \theta \leq 1$, and $\sum_i \theta_i = 1$. (Statement of site balance)

Let's test our assumption that step 3 is the rate limiting step. We need:

expressions for $\theta_{O_2} \approx \theta_*$.

It is difficult to measure the coverages of catalytic species under reaction conditions, so instead, we need expressions for them.

write expressions using the remaining steps
(1, 2, 4, 5) in the mechanism.

θ_{O_2} : get that from step 2

$$0 = r_{net,2} = r_{f2} - r_{r2} = k_{f2} C_{O_2(g)} \theta_* - k_{r2} \theta_{O_2}$$

$$\theta_{O_2} = K_2 C_{O_2} \theta_*$$

neglect since $\theta_{O_2} \ll \theta_*$ neglect since $\theta_{NO_2} \ll \theta_*$

θ_* : site balance: $\theta_* = 1 - \theta_{NO} - \theta_{O_2} - \theta_0 - \theta_{NO_2}$

θ_{NO} : step 1: $\theta_{NO} = K_1 C_{NO} \theta_*$

θ_{NO_2} : step 5: $\theta_{NO_2} = \frac{C_{NO_2} \theta_*}{K_5}$ ← neglect

θ_0 : step 4: $\theta_0 = \frac{\theta_{NO_2} \theta_*}{K_4 \theta_{NO}} = \frac{C_{NO_2} \theta_*}{K_4 K_5 K_1 C_{NO}}$ or step 3, if step 4 is chosen as the RDS

$$\theta_* = 1 - K_2 C_{O_2} \theta_* - K_1 C_{NO} \theta_* - \frac{C_{NO_2} \theta_*}{K_5} - \frac{C_{NO_2} \theta_*}{K_4 K_5 K_1 C_{NO}}$$

$$\theta^* = \frac{1}{1 + \cancel{K_2 C_{O_2}} + K_1 C_{NO} + \frac{C_{NO_2}}{K_4 K_5 K_1 C_{NO}} + \frac{\cancel{C_{NO_2}}}{\cancel{K_5}}}$$

- rewrite rate expression $(-r_{NO})$:

$$K_3 K_2 C_{O_2}$$

$$-r_{NO} = \frac{(1 + \cancel{K_2 C_{O_2}} + K_1 C_{NO} + \frac{C_{NO_2}}{K_4 K_5 K_1 C_{NO}} + \frac{\cancel{C_{NO_2}}}{\cancel{K_5}})^2}{\begin{matrix} \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ \frac{\theta^*}{\theta^*} & \frac{\theta_{O_2}}{\theta^*} & \frac{\theta_{NO}}{\theta^*} & \frac{\theta_0}{\theta^*} & \frac{\theta_{NO_2}}{\theta^*} \end{matrix}}$$

so neglect since we're assuming $\ll \theta^*$

- assume:

① $\theta_{O_2} \approx 0$, actually $\theta_{O_2} \ll \theta^*$

② $\theta_{NO} \approx 0$, $\theta_{NO} \ll \theta^*$

③ $\theta_{NO_2} \approx 0$, $\theta_{NO_2} \ll \theta^*$

④ θ^* is the most abundant surface species
- often we assume one MASS $\frac{1}{2}$ that θ of
all other species ≈ 0 .

⑤ $\theta_o \approx 1 \rightarrow \theta_o \theta^* \approx 0$

- re-write θ_o - $r_{NO} = \frac{k_3 k_2 C_{O_2}}{\left(\frac{C_{NO_2}^2}{K_5^2 K_4^2 K_1^2 C_{NO}^2} \right)}$

simplify $\rightarrow -r_{NO} = \frac{k C_{O_2} C_{NO}^2}{C_{NO_2}^2}$ where
 $K = k_3 k_2 K_5^2 K_4^2 K_1^2$
 $\frac{1}{K}$ is a rate constant

- for the full story, Mulla in Catalysis letters 2005